FINAL PROJECT REPORT

Global Fuel Prices and Taxation Trends (2020–2026)

Interactive data analysis using Tableau Public, HTML, Bootstrap and Flask

Project domain: Data Analytics / Energy Economics

Analysis period: 2020–2026, subject to source coverage

Prepared by: Add student/team name and enrollment number

Institution: Medicaps University, Indore (edit if required)

A detailed report of the problem, objectives, dataset, methodology, dashboard design, implementation, testing, findings, limitations and future scope.

1. Executive Summary

Fuel prices affect household expenditure, transport, freight, manufacturing, agriculture and public finance. International comparisons are challenging because prices differ by currency, unit, tax treatment, fuel specification and observation date.

This project, **Global Fuel Prices and Taxation Trends (2020–2026)**, uses Tableau Public to present country-level price comparisons and changes over time. The workbook contains nine worksheets, a dashboard and a story. A website built with HTML, a Bootstrap-based template and Flask provides access to the published views.

Objectives: present available fuel-price information clearly; compare countries; explore trends from 2020 to 2026; examine tax-related indicators where consistently defined data exists; and document the workflow, testing, limitations and future improvements.

Deliverables: nine Tableau worksheets, an interactive dashboard, a Tableau story, a web interface, project documentation and a demonstration video.

Evidence note: exact numeric findings must be verified against the current workbook and source dataset. This report does not invent country averages, tax rates or statistical results not provided here.

2. Introduction and Background

Fuel is a strategically important commodity linking energy markets with everyday mobility, logistics and production. Retail prices may differ across countries and change over time due to a combination of international crude-oil prices, refining and distribution costs, exchange rates, domestic pricing rules, subsidies, excise duties, value-added taxes and other levies.

The 2020–2026 period is a useful window for examining market movement, but visual patterns must be distinguished from verified explanations. A chart can reveal when prices changed; establishing why they changed usually requires additional evidence.

Problem statement: fuel-price information is distributed across datasets and reports. Different units, currencies, periods and tax definitions make direct comparison difficult. This project organizes available information into interactive charts and a guided story.

Scope: countries and dates represented in the underlying dataset; trend analysis, country comparisons, rankings and tax comparisons where suitable fields exist; Tableau publishing and web integration.

Out of scope: the project is descriptive analytics, not a validated forecast or causal policy evaluation. Predictive or causal claims require additional variables and validation.

3. Objectives and Analytical Questions

The project is organized around specific questions so that each chart has a clear purpose and can be evaluated against the available data.

Primary questions: How do recorded prices differ among countries? Which countries rank higher or lower for a selected metric? What increases, decreases or stable periods are visible between 2020 and 2026? Where suitable tax data exists, how do tax measures compare with retail prices? Are there gaps or inconsistent units that may distort interpretation?

Functional objectives: provide clear titles, units, dates and country labels; enable movement from headline indicators to detailed comparisons; use filters where supported; provide separate dashboard and story links; and keep the web interface readable on desktop and mobile.

Success criteria: date and country coverage accurately reflect the source; charts are readable; filters behave consistently; both published views open; website navigation reaches the correct views; and documentation records methods, assumptions and limitations.

Reporting discipline: every claim should identify the measure, period and unit. Rankings should always state the metric and date range used.

4. Dataset and Data Preparation

Reliable conclusions depend on the quality and structure of the source dataset. Fields should be understood before charting: country identifier, observation date, fuel type, price, currency, unit and any tax-related measure. If multiple sources are combined, differences in definitions must be documented.

Recommended fields: country or region; date/year; fuel type; retail price with currency and unit; tax amount, rate or share where available; source name; retrieval date; update frequency; and notes about coverage.

Preparation workflow: inspect data types and date coverage; standardize country names and date formats; check duplicates, missing values and unusual values; document exclusions and conversions; then connect the cleaned data to Tableau and confirm each measure's aggregation.

Comparability safeguards: nominal prices in different currencies should not be treated as a common scale. Currency conversion requires an identified exchange-rate source and date. A common currency does not adjust for purchasing power. Tax amount, tax rate and tax share are different indicators.

Coverage caveat: identify the actual dataset title, publisher, retrieval date and completeness. If 2026 data covers only part of the year, charts and conclusions must disclose this. Missing observations must not be silently treated as zero.

5. Methodology and Analytical Approach

The project follows a descriptive analytics workflow: define questions, inspect and prepare data, select visual encodings, create worksheets, assemble a dashboard, develop a story and validate the published output. This approach helps users inspect patterns without claiming more than the data supports.

Descriptive summaries: use averages or latest-period values with aggregation clearly labelled. **Time-series analysis:** plot observations chronologically to inspect trends and volatility. **Cross-country comparison:** compare countries using consistent units and a defined period. **Tax comparison:** compare price and tax measures only when definitions and periods align.

Aggregation: an unweighted average gives each country equal weight. Population-weighted or consumption-weighted averages answer different questions. KPI labels and captions should state the aggregation method; averages can conceal variation.

Visualization principles: use lines for time trends and bars for categories; keep axes comparable where units match; use color purposefully; sort rankings intentionally; use tooltips for details; and explain conversions or normalized values.

Validation: compare a sample of Tableau aggregates with independent calculations. Confirm filters update the intended worksheets and that the published version matches the saved workbook.

6. Tableau Worksheets and Dashboard Design

The workbook uses nine worksheets as modular analytical components, a consolidated dashboard and a separate story. Worksheets answer individual questions; the dashboard supports exploration; the story arranges selected views in a guided sequence.

Worksheet roles: KPI cards summarize selected headline measures; a trend chart shows movement over time; country comparisons show differences; ranking views highlight upper or lower values; tax-related views compare tax indicators where supported; and detail/distribution views expose country-level variation.

KPI design: each card should show the metric definition, unit and selected period. **Trend design:** use a chronological axis and disclose incomplete periods. **Ranking design:** state the ranking measure and period. **Tax design:** avoid implying causation from a simple association.

Dashboard layout: place headline indicators near the top, followed by trend and country-comparison charts. Place detailed or tax-related views below or alongside them depending on screen size. Consistent spacing, alignment, typography and filter placement improve hierarchy.

Story sequence: begin with the overall trend, move to differences among countries, explore tax-related comparisons where data supports them, and finish with interpretation and limitations. Each story point should answer one question and use a concise caption.

7. Website Integration and Deployment

The published Tableau views are integrated into a web page using HTML and a Bootstrap-based template. Flask provides a lightweight framework for serving pages or organizing routes. The website is the presentation layer; Tableau Public renders the interactive visualizations.

Architecture: browser → Flask route or static page → embedded Tableau Public view → published workbook. Dashboard and story are exposed through separate links or page sections. This separation lets website content change without rebuilding charts, provided Tableau URLs remain valid.

Dashboard: Open Tableau dashboard
Story: Open Tableau story
Demo video: Watch project demonstration

Embedding checks: preserve the official view path; ensure the placeholder has valid width and height; avoid duplicate scripts; open each Tableau URL directly before debugging HTML; inspect browser console/network errors; and test the hosted site on desktop and mobile.

If a view fails when opened directly, investigate publication, permissions or Tableau service availability before assuming the website code is at fault.

8. Findings and Interpretation Framework

The dashboard is intended to reveal trend, country-comparison and tax-related patterns. Exact conclusions must be checked against the live workbook and dataset before submission. This section provides a disciplined framework for reporting findings without mistaking visual association for a proven explanation.

Trend interpretation: inspect increases, decreases, volatility or stable periods. Check whether movements appear across many countries or only a subset. Consider missing months, changes in country coverage and incomplete 2026 observations. A finding should identify the period, metric and unit.

Country comparison: use comparison and ranking views to identify differences in the selected measure. State top or bottom countries only after confirming the filter and period. Nominal price is not a direct measure of affordability because income and purchasing power differ.

Tax interpretation: where tax data exists, examine whether tax measures and retail prices move together. This is descriptive, not proof that taxation alone caused the difference. Crude-oil costs, refining, distribution, exchange rates, subsidies, price controls and market structure may also matter.

Recommended finding format: observation (what pattern appears); evidence (chart, filters, unit and verified values); interpretation (what it may suggest); and caveat (missing data, currency, weighting or alternative explanations). Insert two or three verified findings before submission.

9. Testing and Quality Assurance

Testing covers analytical correctness and web reliability. A visualization can look polished while containing incorrect aggregation, stale data, broken filters or a failed embed. The final published version should be checked.

Data validation: compare sample records and aggregates with the source data; values should agree within documented rounding. **Date coverage:** inspect earliest/latest dates and completeness of 2026; period labels should match available records.

Filter testing: change country, year or fuel type where available and confirm the relevant views update. **Ranking testing:** change period or metric and verify the sort order. **Embed testing:** open dashboard and story directly, then through the website, confirming that both render without persistent loading.

Responsive testing: inspect desktop and mobile widths; charts, labels and navigation should remain usable. **Navigation testing:** follow all links and test browser back navigation.

Operational risks: Tableau Public may be temporarily unavailable; incorrect view paths or permissions can block rendering; insufficient embed height can make a chart appear blank; and data refreshes can change values. Record a data retrieval date and rerun tests after updates.

Readiness rule: data checks pass, both views open directly, both embeds work through the website, and report claims match the live visualizations.

10. Limitations, Ethics and Future Scope

Limitations: source accuracy and update frequency constrain conclusions; country and month coverage may vary; currency, units, fuel specifications, tax definitions and pricing rules differ; descriptive charts do not prove causation or forecast prices; averages can conceal local variation; and Tableau Public embeds depend on external availability.

Responsible interpretation: use neutral language when discussing country rankings. Display units and periods, avoid selectively omitting observations, distinguish evidence from hypotheses, and never silently treat missing values as zero. Use accessible contrast and meaningful titles.

Future enhancements: add a source panel with dataset title, publisher, retrieval date and definitions; include inflation-adjusted or purchasing-power-aware comparisons where defensible; separate gasoline and diesel analysis if data permits; distinguish tax amount, tax rate and tax share; automate refresh and validation; and add accessible text summaries and mobile-specific improvements.

Statistical modelling should be considered only after assembling appropriate explanatory variables and validating assumptions. A forecast should be clearly separated from historical descriptive analysis.

Maintenance plan: archive source data, refresh the workbook, rerun validation, review changed values, republish views, test embeds and update the report cutoff. Keep workbook and website source files under version control.

11. Conclusion, References and Submission Links

The Global Fuel Prices and Taxation Trends (2020–2026) project demonstrates how interactive visualization can make a complex multi-country energy topic easier to explore. Tableau worksheets provide modular views, the dashboard supports comparison, and the story guides viewers through selected questions. The web interface gives users a unified entry point to the published visualizations.

The project's value depends on transparent preparation, clear units and appropriately qualified interpretation. Country price differences can be explored and tax measures compared where consistent data exists, but causal conclusions require additional evidence. Before submission, verify numeric findings and ensure the workbook, website and report agree.

Dashboard: Global Fuel Prices and Taxation Trends Dashboard
Story: Global Fuel Prices Story
Demo video: Project demonstration video

References: add the exact underlying dataset title, publisher, URL, retrieval date and license. Supporting platform documentation: Tableau Public (https://public.tableau.com/); Tableau Help (https://help.tableau.com/); Flask (https://flask.palletsprojects.com/); Bootstrap (https://getbootstrap.com/).

Final checklist: add student/team name, enrollment number, faculty guide and submission date; enter the actual data source and retrieval date; insert verified numeric findings; test all links on another device; and confirm embedded views load before submission.